

Notes on Susan Leigh Star's 'The Ethnography of Infrastructure'

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As part of the [COPIM project](#), my work packages are conducting some background reading groups. This week we are reading Susan Leigh Star, 'The Ethnography of Infrastructure', *American Behavioral Scientist*, 43.3 (1999), 377–91

[<https://doi.org/10.1177/00027649921955326>](https://doi.org/10.1177/00027649921955326). I had read this a long time ago but enjoyed revisiting it. I thought, in a spirit of openness, that I would share my notes on this article.

This article starts with a humorous note, namely that infrastructure is mundane. This article is, therefore, 'a call to study boring things' (377). The first examples that Star gives are the International Classification of Diseases (ICD) and the telephone book. Star claims that the latter is potentially more interesting; a telephone book tells you a great deal about the demographics of an area through how businesses present themselves, restaurants and ethnicity etc. I think this is disingenuous, though. ICD is important *and interesting* because it shows precisely what is considered a disease/illness and what is thought outside of that purview. Given the degree of controversy around medical diagnoses of psychiatric complaints and the Diagnostic and Statistical Manual of Mental Disorders (DSM), Star knows that this example is not really fair; the ICD tells us a lot, too, about that most socially studied of fields: medicine.

There are challenges in studying infrastructure using conventional ethnographic principles. Not least of these is the scale of the undertaking. Ethnography is very good at focusing on the unique and the specific and surfacing these micro-narratives. Yet what do you do when the object of your study is massive? Yet we need to study infrastructure. Literally, in etymological terms, we are talking about an under (infra-) structure; that which undergirds other things. If you do not focus on these underpinnings, you ‘miss essential aspects of distributional justice and planning power’ (379).

What are infrastructures? For Star, infrastructures are always relational – that is, they must exist in relation to a specific actor. A staircase is one person’s infrastructure – seamless, invisible, functional – but for the wheelchair user, they are a barrier (380). Star gives a series of characteristics of infrastructures (381–2). They have:

- Embeddedness: ‘sunk into and inside of other structures’;
- Transparency: it ‘invisibly supports’ tasks;
- Reach or scope: infrastructure ‘has reach beyond a single event or one-site practice’;
- Learned as part of membership: communities of practice take their infrastructures for granted;
- Links with conventions of practice: there is a type of path dependency on past conventions. Star gives the example of the QWERTY keyboard;
- Embodiment of standards: infrastructures work with other infrastructures through common interfaces;
- Built on an installed base: ‘Infrastructure does not grow *de novo*; it wrestles with the inertia of the installed base and inherits strengths and limitations from that base’;
- Becomes visible upon breakdown: the invisibility of infrastructure fades away when it breaks;
- Is fixed in modular increments: infrastructures are not installed *in toto* in one go.

Of particular interest to us, I think, is the way in which scholarly communication infrastructures have broken down in the current COVID-19 crisis. At a time when we cannot access physical libraries and become dependent on offsite access (which, I should add, is actually many people’s

experience anyway if they are disabled, for instance), the infrastructure is revealed in its brokenness to those who had previously taken it for granted as part of their membership of an academic community of practice.

How should we study infrastructure? There are several ‘tricks of the trade’ (384–7) to which Star points us:

- Identifying master narratives and ‘Others’: basically, we can ask about the assumptions that any infrastructure makes. A good example that Star gives is the idea of labelling sticking plasters as ‘flesh colored’ (385), but that this actually means white skin;
- Surfacing invisible work: invisible labour pervades most infrastructures and, as with the breakdown of the infrastructure as a whole, is only usually noticed when it fails;
- Finding paradoxes of infrastructure: the example Star gives is that small technical challenges – such as adding a single extra button in a workflow – present near-insurmountable technical challenges to users (387–8). This is because of the ways that components interact in assemblage: for instance, a user has a memory of workflow.

The second of these points has a strong resonance for me because, for Star, there is a delicate line in making tacit work explicit. Star uses the example of nurses. If their work is left unnoticed, it ‘fades into the wallpaper’. However, if you make it visible, ‘it will become a target for hospital cost accounting’ (386). Consider the parallel here with Learned Societies. These organizations are currently often funded through the indirect channel of subscription publications. And they like it this way, because it is very difficult for a manager to defund them – to make them the target of cost accounting. However, it also comes with a concomitant difficulty in articulating the value of what such Societies do. Would we pay for Learned Societies if they were forced to make the direct pitch for their value to universities, rather than cloaking the labour and cost beneath resource purchases in cross-subsidy? One could also consider the types of hidden labour at university presses, where attempts have been made to say what an acquisitions editor actually adds, for instance. However, this comes with the same set of risks (and I do now know of at least one major figure in the US university press scene who articulates this value as ‘infrastructure’).

So how do we read infrastructure? Star suggests that we must read across several different levels (387–8):

1. As a constructed material artifact: ‘with physical properties and pragmatic properties in its effects on human organization’;
2. As a trace of record of activities: as a type of imprint or ‘information-collecting device’;
3. As a representation of the world: a kind of model with a ‘sort of substitution’ at play.

Star notes that these indicators, as she calls them, have parallels elsewhere. She uses the distressing example of films that include depictions of rape, which may tell us much about attitudes towards rape in a society, but are not the same as police statistics about rape or phenomenological investigations into the experience of being raped (388).

Star closes with some observations on how infrastructures can encode inequalities, using the example of a town planner who deliberately built low bridges to prevent those using buses from entering certain parts of a city. (It is not clear whether Star takes this at face value as she implies that it is possible to be sceptical about it on page 389.) However, she notes in closing that applying the study of ethnography ‘to this class of issues is a terrifying and delightful challenge for what some would call the information age’.

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